

Research

Hypertension (high blood pressure) is a major health concern in Singapore. According to the National Population Health Survey (NPHS) 2024, about 33.8% of Singapore residents aged 18-74 have hypertension, meaning roughly 1 in 3 adults are affected.

- More than half (51.2%) of hypertensive individuals were previously undiagnosed.
- 60.4% had poor control of blood pressure
- Prevalence for hypertension increases with age, starting at 7.1% for 18 to 29 years old, reaching 69.4% for those aged 70 to 74 years old.

Blood pressure is currently measured using blood pressure monitor equipment. Prices can range from [\\$30 to \\$200](#) (Amazon, n.d.). This can make it difficult to keep track of blood pressure in day to day life, especially for those who do not have the financial means and knowledge to operate the equipment.

Target Audience

Adults ages 40 and above in Singapore with early or diagnosed hypertension.

Justification from NPHS data:

- Hypertension prevalence rises sharply after age 40.
- Many individuals are undiagnosed or poorly managing their condition.
- Lifestyle factors such as high sodium intake and sedentary lifestyles contribute to risk.

Methods to support selected audience

Proposed mobile app: **No Pressure**

A mobile health companion app designed to help users track and improve their blood pressure through lifestyle guidance and reminders.

Features include: Step counter, diet management, personalised risk score to inform users on how susceptible they are, calendar with medicine schedule and reminders that can be set up by the doctor.

Ethical Conflicts

There is a major conflict of Data Privacy vs. Personalization. The app requires sensitive health data (BP readings, medication, and lifestyle habits) to function, mismanagement of this data could lead to a breach of trust or unauthorized profiling.

App login stores user data for personalized schedules and reminders, which may cause concerns about invasion of privacy if mishandled.

Mitigation

Following the Personal Data Protection Act, we provided Terms & Conditions when registering

accounts, being transparent on what data is being used and for what reason while allowing users to drop out anytime.

User persona

Name: Sum Ting Wong

Age: 67

Stats (1 - 5)

Average Daily Blood Pressure: 3 (Low Risk - High Risk)

Memory: 2 (Forgetful - Photographic Memory)

Frequency of exercise: 3 (Lack of exercise - Healthy amount)



Scenario

Wong was diagnosed at the age of 42 with hypertension. Worried about the risk of heart attack, he started exercising and seeing the doctor frequently to manage his health. Recently he has noticed his failing memory and old age causing difficulties in managing his healthy lifestyle. This includes missing appointments, forgetting to take medicine, and difficulties with managing diet and exercise.

App Use Case steps

1. Wong logs into the app around dinner time using their username and password.
2. After logging in, he is taken to the dashboard, where they can view their blood pressure control status, step progress, diet progress, and calendar summary.
3. Around dinner time, he opens the Diet Manager page to monitor their daily sodium intake. He scanned the food he is about to eat, which the sodium content is determined to be 800mg. This brought his total intake to 90% of the safe amount as he only had breakfast today.
4. The user returns to the dashboard to review their daily health progress. The overall blood pressure control reads 65%, warning Wong that he is not managing his health well.
5. Because of this, he first opens the Step Counter page to check their daily steps and review their step count history. Compared to previous days, he did not keep up with his exercise today and decided to take a brisk walk outside.
6. Some time after closing the app, the Calendar feature reminds him about important events created by the doctor such as medication timings or routine appointments. He is given a reminder in advance about a medical appointment tomorrow.

References

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